Spotlight and handover: Register socialization in Korean child-directed speech

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Abstract

How do children learn to map abstract social categories onto complex linguistic signals? While developmental research has benchmarked *when* children recognize social registers (e.g., matching casual vs. formal registers to specific interlocutors), the *mechanisms* by which they acquire these context-dependent mappings remain unclear. We leverage the high-resolution morphological signal of Korean honorifics to investigate the dynamic processes underlying early register socialization. Through analysis of cross-sectional and longitudinal transcripts from CHILDES, we demonstrate that caregivers spotlight the socio-pragmatic functions of honorifics in child-directed speech (CDS) by selectively clustering them within specific communicative acts. Furthermore, we reveal a pragmatic handover, in which caregiver scaffolding systematically fades as the child's independent sociolinguistic competence emerges. The findings suggest that register learning is supported by an adaptive tutoring process facilitating the child's transition from a linguistic recipient to a register-flexible social agent.

Keywords: language input; child-directed speech; register; honorifics; linguistic socialization; social deixis

Introduction

A crucial part of becoming an expert language user is learning to flexibly adapt speech to different social contexts: a process of *register learning*. Unlike word learning, where a discrete linguistic form maps to a referent, registers consist of clusters of probabilistic linguistic and metalinguistic features across an utterance. These may include specific lexical forms (e.g., *please*), syntactic structures (e.g., indirect requests; *could you pass the salt?*), alternate second person pronominal paradigms (e.g., Spanish *usted* vs. *tu*), and specific prosodic contours (e.g., exaggerated pitch in infant-directed speech (IDS); Hilton et al., 2020; Kuhl et al., 1997; Piazza et al., 2017). Register learning thus presents a complex socio-cognitive problem. Children must map multifaceted linguistic signals to the abstract social dimensions of hierarchy, distance, and power. Critically, the perceptual saliency and structural resolution of these signals vary across sociolinguistic ecologies, ranging from subtle prosodic shifts in some languages to obligatory morphosyntactic markers in others. Here, we investigate the developmental origins of register knowledge by focusing on Korean honorifics: a collection of register-like speech styles marked in high-resolution thanks to alternate vocative forms, morphosyntactic inflections, lexical substitutions, and more. Our work shifts the focus of register learning research from behavioral markers of comprehension to the dynamic structure of the linguistic input itself to examine register learning in situ.

Work by Wagner et al. (2010) suggests that, by age 5, children correctly match different registers (IDS, Casual, Formal) with socially-appropriate conversational partners (baby, little girl, teacher). By this age, children also provide sophisticated metalinguistic justifications for their choices (e.g., "because people say 'hi' to kids"). Notably, children learning languages with more explicit grammatical cues to register (e.g., Spanish) seemed to have greater success in these socio-pragmatic tasks than their English-learning counterparts (Wagner et al., 2014).

Beyond speech categorization, register knowledge carries profound socio-cognitive implications. Children demonstrate social biases toward register-competent speakers as early as age 4 (Ambrish et al., 2025), a preference that matures by age 7 into a tool for epistemic evaluation where register accuracy serves as an index of credibility (Ikeda et al., 2019). Thus, register is treated by young children as not merely as a set of linguistic rules, but rather a normative social code used to evaluate the competence and reliability of conversational partners. In this view, register errors are seen as social transgressions and markers of unreliability that fundamentally shape a child's social and learning landscape.

However, a fundamental theoretical challenge remains: *how* do children decode these complex social registers from their linguistic environment? Children have access to register information through multiple pathways. Because children are rarely the targets of formal address in many cultures, it is often assumed that register knowledge develops primarily through overheard third-party interactions. Indeed, recent work in high-density social environments has demonstrated that infants can acquire honorific vocative-referent mappings through overheard speech alone, even when directed input is sparse (Foushee & Srinivasan, 2024).

Yet, overheard speech may represent only one source of register input. Caregivers also make active efforts toward linguistic and behavioral socialization through child-directed speech (CDS; Ochs and Schieffelin, 1986). For example, caregivers frequently adapt their registers in CDS to mark specific socio-pragmatic goals, such as using stern affective contours to accompany behavioral directives (Fernald, 1989). We propose that caregivers complement overhearing-based learning by providing a dynamic socio-pragmatic framework in di-

rected input. In this view, caregivers *spotlight* the functions of the formal register by selectively deploying it within communicative acts most conducive to politeness. Furthermore, we suggest this scaffold is not used statically over development; rather, it exhibits a *pragmatic handover* (Bruner, 1983), where the density of directed register input is dynamically calibrated and eventually faded as the child's independent mastery emerges. To uncover the cognitive mechanisms of register acquisition, we must look to "high-resolution" languages where social register is not merely an optional stylistic variation, but is consistently and obligatorily encoded through grammatical markers.

Korean provides an ideal testing ground for these questions. While register remains a contested concept, we follow Brown (2015) in treating the Korean honorific system—and specifically its "speech styles" (*chey*)—as a functional register framework. These styles represent distinct, situation-bound varieties that index the social relationship between interlocutors across almost every utterance. This system encodes social deixis (Levinson, 1979) through a complex, compositional register of lexical alternates (e.g., *saengil* vs. *saengshin* for *birthday*) and morphosyntactic inflections (e.g., the verbal suffix *-yo*).

We argue that the morphological affordances of this system—specifically its regular usage and its position as an obligatory, agglutinated verbal suffix—provide a salient and reliable signal for the young learner. Unlike the subtle prosodic or metalinguistic cues of English, the Korean honorific suffix offers a tag that is structurally discrete and perceptually accessible, allowing children to align linguistic cues to their social understanding with high precision. Crucially, recent work has noted that, contrary to the expectation that formal registers are only overheard, Korean caregivers utilize honorifics in CDS (Ahn, 2020; Bae et al., 2024). This usage supports our hypothesis that caregivers actively provide a linguistic scaffold for social deixis, though the systematic nature of this input calibration remains unexplored.

Current Investigation

We utilize cross-sectional and longitudinal data from naturalistic corpora to investigate the developmental landscape of Korean honorifics, tracking both the temporal trajectory of caregiver input (*when*) and the specific socio-pragmatic contexts in which this input is situated (*how*). In Study 1, we analyze mother-child play sessions from 30 dyads at three developmental milestones (0;8, 1;1, and 2;3) to establish how caregivers spotlight social information. By coding and analyzing over 30,000 utterances for specific speech act categories, we reveal that honorific usage in CDS is not a static marker of social distance, but is selectively clustered to highlight socio-pragmatic functions.

In Study 2, we explore the longitudinal dynamics of this learning environment by examining the extent to which caregiver input is calibrated to the child's emerging competence. An analysis of over 103,000 spontaneous utterances across three families reveals a consistent pragmatic handover in the honorific system; by age three, the child's increasing use of honorific registers begins to converge with declining honorific density in caregiver speech. While family-specific baselines vary significantly in their overall frequency of honorific use, the developmental slope of this handover remains remarkably uniform across families.

Together, these studies take a first step toward understanding register learning as a dynamic process of socio-pragmatic calibration. By leveraging the structural affordances of honorifics, Korean caregivers transform an abstract social code into a salient pedagogical signal. This spotlighting enables children to map linguistic variants onto social dimensions of hierarchy and grounding. Ultimately, we argue that honorifics in child-directed speech function as dynamic scaffolds of social information, facilitating the child's transition from a linguistic recipient to a register-flexible social agent.

Study 1: Honorifics in the spotlight

We propose that the use of honorifics in Korean CDS is a nuanced and dynamic tool for register socialization. To establish this pedagogical function, we contrast the observed patterns of honorific usage against two alternate, more static frameworks that might otherwise account for their presence in the input.

First, we contrast our scaffolding account with the possibility that honorifics are merely *static markers of social distance*. If honorific usage were strictly a reflection of the intimacy of a relationship, caregivers should exhibit a uniform baseline of politeness across all interactions with family members, regardless of the addressee. Second, we test whether honorifics function as *general markers of linguistic input*, mirroring the raw frequency of different speech acts.

Through these comparisons, we demonstrate that honorifics are neither a static reflection of the dyad nor a proportional byproduct of frequency. Instead, our speech act analyses suggest that caregivers utilize the Korean honorific register as a dynamic socio-pragmatic signal, spotlighting the register within specific communicative contexts to orient the child toward its social significance.

Method

Corpus

We analyzed transcripts and audio from the Ko corpus (Ko et al., 2020), which is available via the Child Language Data Exchange System (CHILDES) (MacWhinney, 2000). The corpus consists of cross-sectional recordings of 30 mother-child dyads residing in the Seoul metropolitan area, divided into three developmental groups: 8 months ("Preverbal"), 13 months ("Early speech"), and 27 months ("Multi-word"). Ten dyads were included per age group.

Recordings took place in a university-based mock apartment and consisted primarily of spontaneous mother-child play (mean duration = 46.87 min; *SD* = 3.44). Crucially, the sessions also captured mother-experimenter interactions and 3–5 minute phone calls to family members, providing a

naturalistic contrast between CDS and adult-directed speech (ADS) within the same recording session.

Speech act coding

We use the Inventory of Communicative Acts - Abridged (INCA-A) (Ninio et al., 1994) to code for speech acts at the utterance level. Using the turn-constructional unit (TCU) as the basic unit of analysis (Sacks et al., 1974), each utterance was assigned one of 67 illocutionary force codes, spanning across 12 sub-categories (Directives, Speech elicitations, Commitments, Declarations, Markings, Statements, Questions, Performances, Evaluations, Demands for clarification, Text editing, and Vocalizations). All coding was performed by the first author, a native Korean speaker.

Register categorization

Utterances were automatically coded for register markers using regular expressions. Following the framework of Brown (2015), we categorized utterances into four primary registers based on their sentence-ending verbal suffixes (*chey*) and lexical alternates:

- **Honorific/Deferential (*hapsyo-chey*):** Marked by high-level formal suffixes, including *-(su)pnita* (declarative) and *-(su)pnikka* (interrogative).
- **Honorific/Polite (*hayyo-chey*):** Representing the standard polite register, this category was identified by the verbal suffixes *-yo* and *-cyo* (or *-jo*), as well as the polite affirmative lexical items *ne* and *ye* ("yes").
- **Authoritative (*hao-chey*):** While traditionally categorized as a distinct formal style, this register is rarely heard in modern spoken Korean and is considered by some to be "headed for extinction" (Brown, 2015). Nevertheless, we included the suffix *-(s)o* (and its imperative form *-shio*) in our initial analysis to ensure the full capture of any hierarchical directives or "frozen" formal routines that might still persist in child-directed input.
- **Non-honorific (*panmal*):** Representing the intimate (*hay-chey*) and plain (*hayla-chey*) styles, identified by suffixes such as *-e/a*, *-ni*, and *-ta*.

Additionally, we also coded for social deixis in the pronominal system, specifically tracking the use of the "humble" first-person pronoun *ce* (and plural *cehuy*), which contrasts with the "plain" forms *na* and *wuri*.

Results

Register composition of Korean CDS

To characterize the baseline linguistic environment, we first analyzed the global distribution of registers in caregiver input. Across all developmental milestones, the casual register (*panmal*) constituted the primary mode of address (8mo: $M = 86.2\%$, $SD = 5.6\%$; 13mo: $M = 87.3\%$, $SD = 5.9\%$; 27mo: $M = 82.6\%$, $SD = 9.0\%$). However, the formal registers remain a robust and regular feature of early child-directed input.

The polite register (*hayyo-chey*) provides the most frequent kind of formal signal, appearing in roughly 10–15% of utterances. Notably, usage of this register exhibits an "Early speech" dip at 13 months ($M = 10.3\%$, $SD = 5.7\%$) before increasing substantially as children enter the multi-word stage at 27 months ($M = 14.7\%$, $SD = 8.4\%$). The deferential register (*hapsyo-chey*), while relatively rare, shows a steady developmental increase, nearly tripling in frequency from 8 months ($M = 0.7\%$, $SD = 0.6\%$) to 27 months ($M = 1.9\%$, $SD = 1.1\%$). Interestingly, the authoritative register (*hao-chey*) is most prevalent in the earliest preverbal stage ($M = 1.2\%$, $SD = 0.6\%$), potentially reflecting its use in "frozen" pedagogical routines or rhythmic modeling for infants, before declining by 27 months ($M = 0.7\%$, $SD = 0.5\%$). Given its overall sparsity, and given its reported trajectory towards extinction, subsequent analyses focus on the polite and deferential registers.

Are honorifics static markers of social distance?

Having established the presence of a formal pedagogical signal CDS, we next evaluated whether these honorifics function merely as static markers of social distance. If honorific usage were strictly a reflection of the intimacy of the social relationship, we would expect caregivers to exhibit a similar baseline of politeness across all interactions with interlocutors of comparable social distance. We thus compare the proportion of honorifics-marked utterances in CDS and other-family-directed speech (OFDS), specifically phone calls to immediate relatives such as spouses, parents, or siblings. One dyad was excluded from this analysis to ensure that all compared interlocutors belonged to a uniform social circle of high intimacy (the excluded mother called a great-aunt, whereas all other interlocutors were members of the nuclear family).

Our analyses underscore that the formal scaffold is a specialized feature of the child-directed environment rather than a general trait of the speaker's register within the intimate domain. Honorifics were used significantly more in CDS than in OFDS ($M = 0.114$ vs. $M = 0.058$; $t(28) = 3.16$, $p < 0.005$). For context, we also drew a comparison to an adult-directed baseline (experimenter-directed speech), which showed a much higher average proportion: 0.786. These results suggest a functional *decoupling* of the register from traditional social hierarchy: caregivers provide a denser formal scaffold to their children—the lowest-status members of the social circle—than to their intimately familiar adults. This finding supports a pedagogical interpretation, wherein honorifics are deployed in CDS to model social registers rather than to index existing social distance.

Are honorifics general markers of linguistic input?

An alternate account for CDS honorifics could be that honorifics are simply general markers of CDS, and as such they are used similarly across speech acts, uniformly tracking change in speech act use over age. However, honorifics usage was not directly proportional to the overall frequency of speech

act categories within age groups ($\chi^2(2) = 60.6; 158.7; 177.6$ for 0;8, 1;1, and 2;3 respectively, all $p < 0.001$). Instead, caregivers utilize the honorific register as a dynamic tool for dual-track socialization: managing the child's behavior while simultaneously scaffolding their emerging linguistic and social knowledge.

The global distribution of the honorific signal (Figure 1) reveals this dual focus. A substantial proportion of the honorifics children encounter are concentrated in categories of behavioral socialization (e.g., Directives and Evaluations; green labels), where the register is used as an extra signal to increase the saliency of commendable behavior. Simultaneously, honorifics occupy a dominant share in categories aimed at linguistic socialization (e.g., Questions and Performances; blue labels), highlighting the register's role in both implicit and explicit modeling of the socio-pragmatic function of honorific registers.

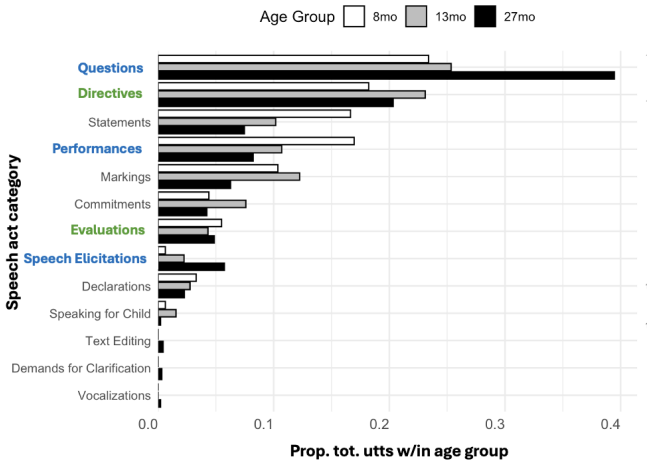

Figure 1: Distribution of honorific utterances across speech act categories per age group.

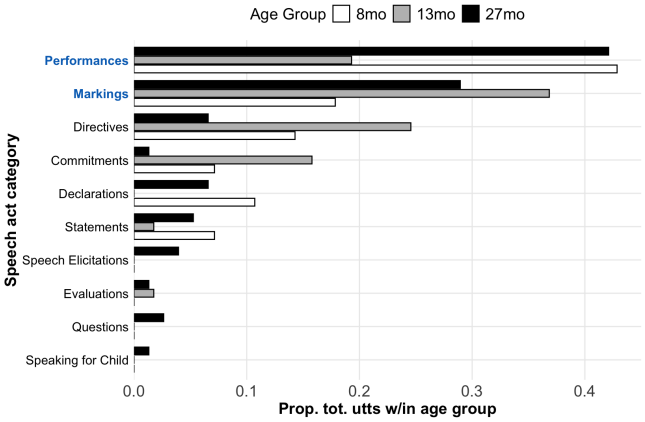

Figure 2: Distribution of deferential honorific utterances across speech act categories per age group.

To further investigate the granularity of this scaffolding, we analyzed the distribution of the most formal register subtype: the deferential register (*hapsyo-chey*). While this register constitutes a small proportion of the overall input (reaching only 1.9% by 27 months), its functional distribution is highly specialized (Figure 2). Note that given the predominance of the polite register, its functional distribution across speech acts closely mirrors these global patterns.

Unlike the polite register, which is broadly distributed across interactional categories like Questions and Directives, the deferential register is overwhelmingly clustered within Performances (e.g., book reading, pretend play) and Markings (e.g., ritualized greetings). For instance, at 8 months and 27 months, nearly 43% of all deferential endings are situated within Performances.

Notably, post-hoc comparisons within a Generalized Linear Mixed-Effects Model (GLMM) with participant as a random intercept revealed that even within the broader honorific registers, the Performances category showed a significant developmental increase in honorific density between 13 and 27 months ($z = -3.33$, $p = .003$). This suggests that caregivers do not use the most formal register for general social interaction, but rather reserve it for specific pedagogical genres. Additionally, caregivers selectively ramp up the formal register in pedagogical and scripted genres as children transition into more complex linguistic production, providing a high-contrast signal that distinguishes scripted social routines from everyday conversation.

Table 1: Description of caregiver CDS across age groups in CHILDES Ko corpus (mean utterances/min, number of unique speech acts, and proportion of honorific utterances).

age group	mean utts/ min (sd)	mean # unique speech acts (sd)	prop. hon. utts
0;8	38.3 (9.4)	29.9 (4.6)	0.085 (0.044)
1;1	39.6 (5.7)	33.4 (3.9)	0.090 (0.057)
2;3	39.1 (3.2)	38.5 (3.1)	0.132 (0.087)

Note also that while usage of honorific registers exhibits a numerical increase as children enter the multi-word stage at 27 months (Table 1, rightmost column), the GLMM revealed that this global cross-sectional shift was not statistically significant ($\chi^2(2) = 3.36$, $p = .186$). The lack of a significant age effect on honorific density may be attributable to high inter-subject variability in caregiver baselines that masks developmental effects in cross-sectional data.

Discussion

Beyond merely marking social distance, our results suggest that honorifics in Korean CDS function as a dynamic scaffold for social register learning. By selectively leveraging the morphological affordances of the system, caregivers spotlight the socio-pragmatic functions of honorifics, orienting chil-

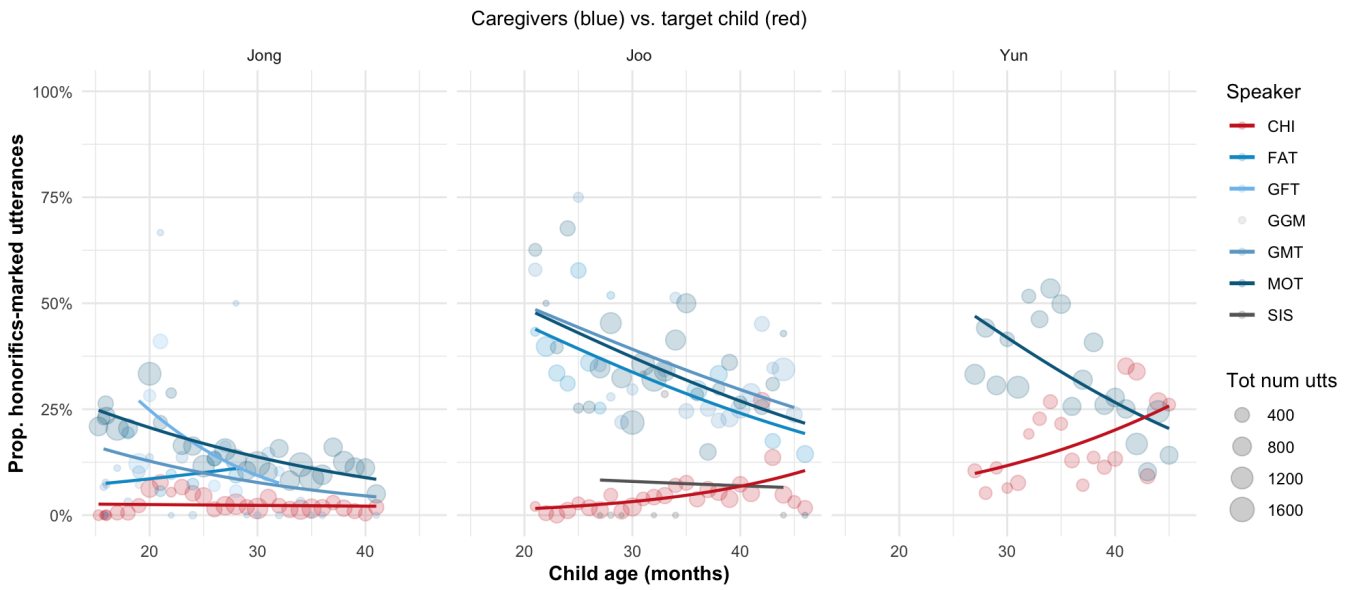

Figure 3: Change in proportion of honorifics-marked utterances by speaker over target child's developmental trajectory.

dren toward their role as politeness markers for information and action requests (Questions and Directives) and as communicative markers for social grounding.

This selective socialization strategy is reflected in the divergent roles of register subtypes: caregivers leverage the polite register for dynamic socio-pragmatic negotiation, whereas the deferential register is reserved for static, rule-governed social routines. By tagging ritualized acts with the most formal register, caregivers provide a high-contrast signal that distinguishes scripted social routines from dynamic, pragmatic usage of registers.

Study 2: Pragmatic handover

While Study 1 established the functional clustering of CDS honorifics, the lack of a global cross-sectional effect (coupled with high inter-subject variability) suggests that individual caregivers operate from distinct politeness baselines. These variable baseline individual differences across age groups make it difficult to detect subtle developmental calibration across different families. We therefore turn to longitudinal data to better isolate how directed register input is dynamically calibrated within individual dyads as the child matures.

Method

Corpus

We analyze the Ryu corpus (Ryu et al., 2015) from the CHILDES database, which contains longitudinal, naturalistic recordings of three Korean children and their families. The dataset spans the critical period of early language acquisition, covering ages from approximately 15 to 46 months, and recording contexts remain roughly consistent across recordings for a child. The corpus consists of over 103,000 spontaneous utterances, including speech from the target children and their primary caregivers (mothers, fathers, sibling, and grandparents).

Statistical analysis

Utterances were automatically coded as in Study 1. We employed a longitudinal GLMM with a binomial link function to predict honorific usage. The model included an interaction between *Age* (centered in months) and *Speaker Role* (Child vs. Caregiver identity), with *Family ID* as a random intercept to account for varying household politeness baselines.

Results

The GLMM revealed a highly significant interaction between age and speaker role ($\beta = 0.104, z = 28.58, p < .001$), supporting the pragmatic handover hypothesis. Caregivers exhibited a robust longitudinal decline in honorific density as children aged ($\beta = -0.045, z = -34.31, p < .001$), reflecting the systematic fading of the pedagogical scaffold as children become more linguistically and socially competent. Conversely, children showed a significant developmental increase in independent honorific production (net slope $\beta = +0.059, p < .001$). This diverging trajectory (Figure 3) demonstrates an emerging register convergence: as the child's sociolinguistic competence grows, the caregiver systematically withdraws the formal scaffold, allowing families to settle into the casual register that traditionally defines the adult-child relationship.

The pragmatic handover is characterized by a reciprocal shift in register density. While caregiver input declines and child production rises across all families, the data reveals an emerging register convergence. In the case of Yun—the family with the highest adult baseline (Figure 3, rightmost panel)—we observe a clear crossover point at approximately

38 months, where the child's honorific density begins to exceed the declining caregiver input. In the families of Jong and Joo (Figure 3, leftmost and center panels), the trajectories are clearly trending toward this intersection, though the handover remains incomplete within the 40- to 46-month window.

These findings suggest that the rate of socialization (here, the slope of change in honorifics usage) is a robust cognitive constant. However, the attainment of mastery (the crossover point) appears to be calibrated to the family's specific politeness set-point. For example, Yun's earlier crossover may reflect a high-pressure socialization environment where the density of the initial scaffolding accelerated the child's independent register adoption.

Discussion

The systematic fading of caregiver honorifics suggests that once the child begins to produce the formal register independently, the caregiver shifts the target of socialization from register modeling to register-flexible interaction, allowing the dyad to settle into the casual register that traditionally defines the parent-child relationship.

We propose that the systematic withdrawal of honorifics is an adaptive response to the child's emerging competence. Rather than waiting for the child to reach full adult-like proficiency, caregivers begin fading the scaffold as the child begins to demonstrate reliable independent production. The fact that the crossover is only fully realized in one child (Yun) within our time frame suggests that the handover is a prolonged developmental process. Caregivers are not necessarily aiming for a specific frequency of honorifics, but are instead handing over the responsibility for social deixis as the child's socio-pragmatic agency emerges.

General Discussion

Our results characterize register learning as a process of socio-cognitive calibration. By synthesizing cross-sectional and longitudinal data, we suggest that register input reflects a deliberate pedagogical intent. Identifying the structural resolution of this input is a necessary precursor to understanding how children transition from formulaic reproduction to social agency.

Beyond routines: Register as pragmatic negotiation

Our functional analysis reveals a core tension between register as a scripted routine and as a tool for flexible pragmatic negotiation. Much of the existing literature has benchmarked register knowledge through the child's ability to navigate greeting sequences and formulaic social routines (Ikeda et al., 2019; Wagner et al., 2014; Wagner et al., 2010). While these formulaic anchors are present, we find that the honorific signal is concentrated in interactional categories that require active social grounding, such as Questions, Directives, and Performances (e.g., role play).

This clustering suggests that caregivers do not merely model honorifics as frozen formulas for ritualized greetings. Instead, they use the honorific register to tag communicative acts that involve information requests and social perspective-taking. By ramping up honorific density within these flexible, pragmatic categories as children transition into the multi-word stage ($p = .003$ for Performances), caregivers provide a training signal that distinguishes rule-governed social performance from dynamic conversation. Standard assessments using greeting-based scenarios may therefore underestimate children's competence, as they overlook the child's emerging ability to use registers to negotiate social agency and intent in real time. It remains to be determined whether early child production demonstrates true sociolinguistic flexibility—such as correct usage with novel interlocutors—or merely mimics frozen forms learned within specific routines. The developmental transition from rote reproduction to context-sensitive deployment marks the achievement of true register-flexible agency.

The handover and universal scaffolding

The longitudinal crossover observed in Study 2 marks the child's transition from a linguistic recipient to an active social agent. In the Yun family, we observed a clear intersection where the child's honorific density began to exceed the declining caregiver scaffold. While the Jong and Joo families showed trajectories still trending toward this intersection at 46 months, the uniformity of the slopes across families suggests a shared cognitive mechanism of adaptive tutoring. Caregivers appear to monitor the child's emerging production in what may function as a closed-loop feedback system, withdrawing the pedagogical support exactly as the child demonstrates independent production.

We expect that comparable scaffolding mechanisms operate across diverse linguistic systems. Even in morphologically sparse languages like English, caregivers likely spotlight social registers through specific prosodic contours or affective shifts during directives and questions. Korean honorifics simply provide a privileged window into this universal architecture.

Conclusion

How do children learn socially-conditioned language use? By utilizing the structural resolution of the Korean honorific system, we move from benchmarking when children recognize social registers to uncovering the adaptive tutoring process that makes such mappings possible. Ultimately, these results reveal a broader cognitive architecture for children's socialization—one where the structure of the environment is dynamically tuned to facilitate the learner's transition from a passive recipient of cultural information into a flexible and normative social agent. Future research should apply this decomposition approach to other modalities—such as prosody and gesture—to test whether caregivers in other cultural groups employ a similarly adaptive pragmatic handover for socially conditioned language use.

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